Course Work Report: Digital Game Distribution Platform (Steam-like Service)

Topic: a Steam-like platform where users buy digital games, add games to wishlist, write reviews, earn achievements and where business analysts track sales and user activity.

DBMS: PostgreSQL. OLTP and OLAP are stored in separate schemas: steam_oltp and steam_dwh.

1. OLTP Database Context

The OLTP database supports operational functionality of the application. It stores accounts, countries, games, developers, publishers, genres, purchases, reviews, achievements, wishlists and game price history. The schema is normalized to 3NF: repeating groups are separated, many-to-many relations use junction tables, and non-key attributes depend only on their table key.

1.1 OLTP Schema

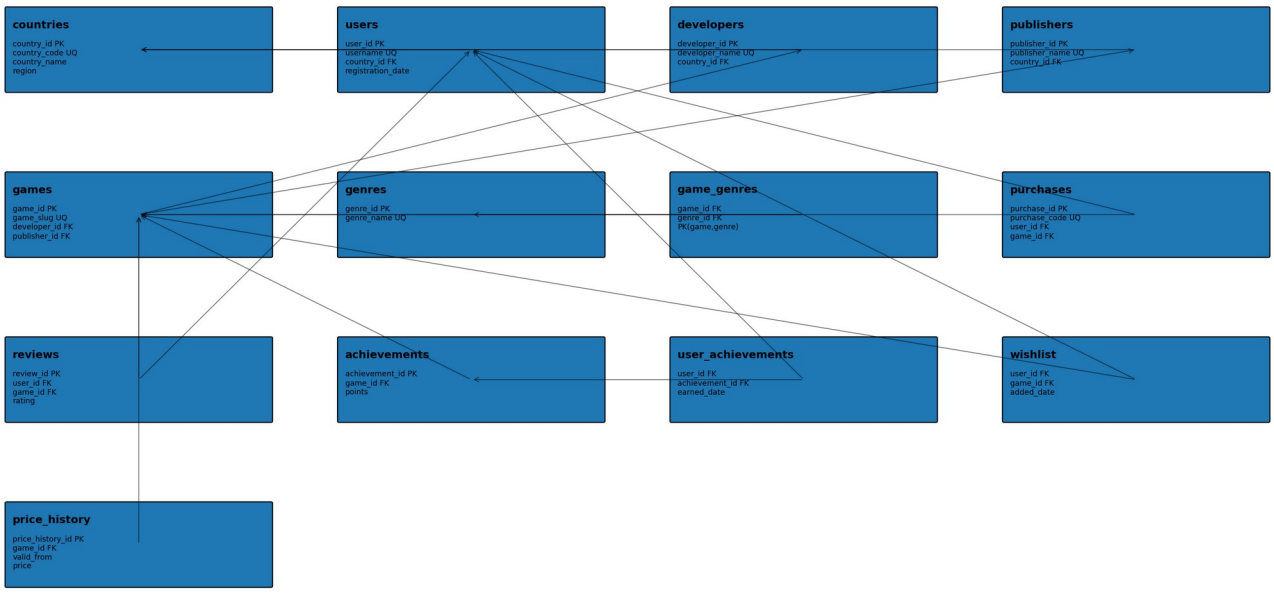


Table	Purpose
countries	Reference table for user, developer and publisher countries.
users	Customer accounts with country and registration date.
developers	Organizations that develop games.
publishers	Organizations that publish games.
games	Game catalog with developer, publisher, release date and base price.
genres	Game genre dictionary.
game_genres	Bridge table between games and genres.
purchases	Game purchase transactions.

reviews	User reviews and ratings.
achievements	Achievements defined for each game.
user_achievements	Achievements earned by users.
wishlist	Games saved by users for future purchase.
price_history	Historical game prices for SCD/analytics.

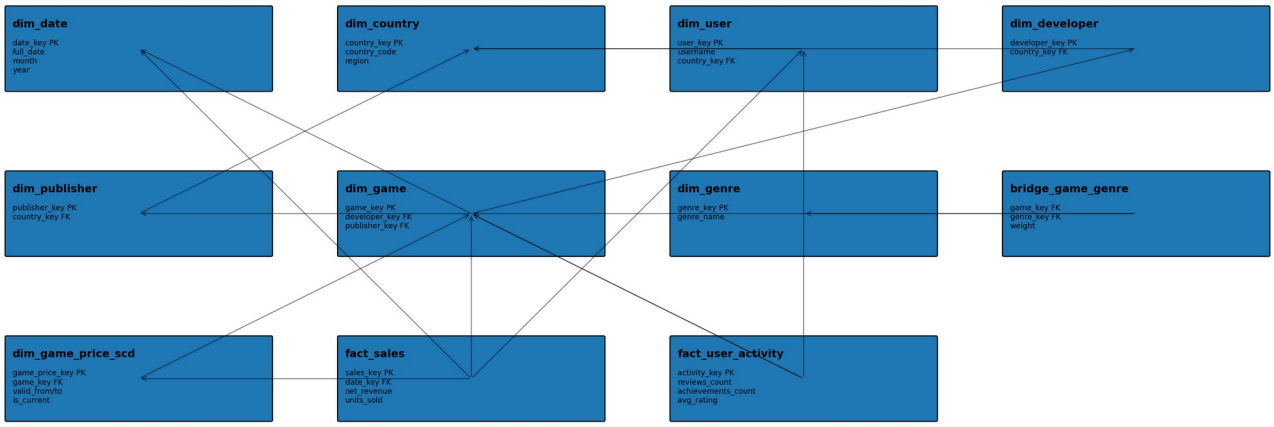
2. Data Loading

Initial data is stored in CSV files. The CSV files do not contain surrogate database keys. Instead, they use business identifiers such as username, game_slug, country_code, purchase_code and achievement_code. The loading script resolves these natural keys into internal foreign keys. The script is rerunnable and uses ON CONFLICT to prevent duplicate data.

3. OLAP Database Context

The OLAP database is a snowflake DWH used to answer analytical questions: which games and genres produce the most revenue, how sales change by month, which countries have the highest purchasing activity, and how reviews/achievements relate to user engagement.

3.1 OLAP Schema



Object	Requirement covered
fact_sales	Fact table 1: revenue, discounts and units sold.
fact_user_activity	Fact table 2: reviews and achievements aggregated by date/user/game.
dim_game_price_scd	SCD Type 2 dimension for game price changes.
bridge_game_genre	Bridge table for many-to-many game-genre analytics.
dim_date, dim_user, dim_game, dim_country, dim_genre	Snowflake dimensions used by the facts.

4. SQL Scripts and Run Order

Order	File	Description
1	sql/01_create_oltp.sql	Creates OLTP schema and tables.
2	sql/02_load_csv_to_oltp.sql	Loads CSV data into OLTP tables. Run from sql folder in psql.
3	sql/03_create_olap.sql	Creates DWH schema and tables.
4	sql/04_etl_oltp_to_olap.sql	Transfers and transforms data from OLTP to OLAP.
5	sql/05_queries_oltp.sql	Operational insight queries.
6	sql/06_queries_olap.sql	Analytical insight queries.

5. Analytical Questions

Question	Where answered
Which games bring the highest revenue?	OLTP query 1 and Power BI bar chart.
What is the average rating for every game?	OLTP query 2 and card/table visual.
How do purchases and revenue change by country and month?	OLTP query 3 and Power BI slicers.
Which genres generate the most allocated revenue?	OLAP query 1 through bridge_game_genre.
How does revenue change by month and year?	OLAP query 2 and line chart.
How do current prices relate to historical sales?	OLAP query 3 using SCD Type 2.

6. Power BI Report Design

Connect Power BI to PostgreSQL schema steam_dwh. Import fact_sales, fact_user_activity, dim_date, dim_game, dim_country, dim_genre, bridge_game_genre and dim_game_price_scd. Recommended visuals:

Visual	Fields
Title	Steam-like Platform Sales Dashboard
Slicer 1	dim_date.year / dim_date.month
Slicer 2	dim_country.country_name or dim_genre.genre_name
Line chart	Axis: Year-Month; Values: SUM(fact_sales.net_revenue)
Bar chart	Axis: dim_game.title; Values: SUM(fact_sales.net_revenue)
Donut/column chart	Axis: dim_genre.genre_name; Values: allocated revenue through bridge_game_genre
Table	Game title, current price, units sold, revenue

7. Defense Notes

The project satisfies the mandatory requirements: OLTP has more than 8 tables, the model is normalized, CSV files use natural keys, the load script is rerunnable, OLAP is separated from OLTP, DWH is snowflake-style, it includes two fact tables, one Type 2 SCD dimension and one bridge table. The analytical model does not duplicate OLTP structure directly because it includes dimensions, facts, aggregations and weighted genre allocation.